Problem 5

Refractive index of water is 4/3 and the refractive index of glass is 3/2. Find the refractive index of glass with respect to the refractive index of water.

Solution

$$ {}_{w}\mu_{g} = \frac{\text{Refractive index of glass}}{\text{Refractive index of water}} $$

$$ = \frac{\frac{3}{2}}{\frac{4}{3}} = \frac{9}{8} = 1.125 $$

3.8.2 Snell's Law of Refraction

Refraction of light rays, as they travel from one medium to another medium, obeys two laws, which are known as Snell's laws of refraction. They are given below:

i) The incident ray, the refracted ray and the normal at the point of intersection, all lie in the same plane.

ii) The ratio of the sine of the angle of incidence (i) to the sine of the angle of refraction (r) is equal to the refractive index of the medium, which is a constant.

$$ \frac{\sin i}{\sin r} = \mu $$

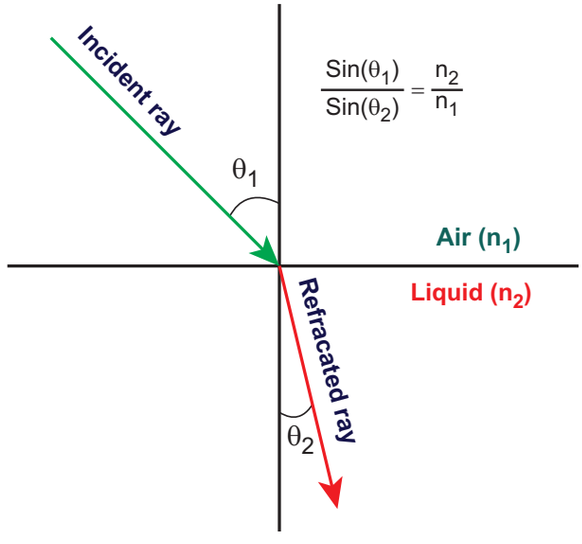


Figure 3.12 Snell's Law

3.9 Dispersion

Activity 7

Place a prism on a table and keep a white screen near it. Now, with the help of a torch, allow white light to pass through the prism. What do you see? You can observe that white light splits into seven colored light rays namely, violet, indigo, blue, green, yellow, orange and red (VIBGYOR) on the screen. Now, place another prism in its inverted position, between the first prism and the screen. Now, what do you observe on the screen? You can observe that white light is coming out of the second prism.



In this activity, you can see that the first prism splits the white light into seven coloured light rays and the second prism recombines them into white light, again. Thus, *it is clear that white light consists of seven colours.* You can also recall Newton's disc experiment, which you studied in standard VII.

Splitting of white light into its seven constituent colours (wavelength), on passing through a transparent medium is known as dispersion of light.

Why does dispersion occur? It is because, light of different colours present in white light have different wavelength and they travel at different speeds in a medium. You know that refraction of a light ray in a medium depends on its speed. As each coloured light has a different speed, the constituent coloured lights are refracted

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